

Name: _____

1. (12 pts) Let $f(x) = (x + 3)(x - 1)(x - 4)$.a. Find all real zeros of f . **Answer:** $x = -3, 1, 4$

b. Determine the following key features:

i. The y-intercept. **Answer:** $f(0) = 12$ ii. Where $f(x) > 0$. **Answer:** $x \in (-3, 1) \cup (4, \infty)$ c. Describe the end behavior of f :i. As $x \rightarrow -\infty$: **Answer:** $f(x) \rightarrow -\infty$ ii. As $x \rightarrow +\infty$: **Answer:** $f(x) \rightarrow +\infty$ **2. (10 pts)** The area of a triangle with sides a and b and included angle θ is $A = \frac{1}{2}ab \sin(\theta)$.1. If $a = 2$, $b = 3$, and θ is increasing at $\frac{d\theta}{dt} = 0.2$ rad/s, find a formula for $\frac{dA}{dt}$ in terms of θ .**Answer:** $\frac{dA}{dt} = \frac{3 \cos(\theta)}{5}$ 2. Find the rate at which the area is changing when $\theta = \pi/3$.**Answer:** $\frac{dA}{dt} = \frac{3}{10} \text{ft}^2 / \text{s}$ 